Unit - 3

Task A

Q.1) What is encapsulation?

Ans⇒ It is the process of hiding data and allowing access only through functions.

Q.2) Diff between inheritance & Composition.

Ans⇒. Inheritance represents "is-a" relationship subclasses can access protected members of the parent.

- Composition represents "has-a" relationship. Components fully encapsulated; the container has no special access to their intends.

Q.3) Why polymorphism is useful in robotics.

Ans⇒ Polymorphism allows the same function to behave differently for different robot components, increasing flexibility.

Task B

Q.1) Identify which relationship is is-a.

Ans⇒ Robot has-a sensor, controller, & Motor - Composition

Q.2) Identify which relationship is is-a.

Ans⇒ A ~~wheel~~ WheledRobot is-a Robot ⇒ inheritance.

Task C:

```
# include <iostream>
using ~~ng~~ namespace std;

class motor {
~~Pinte~~ private:
        int speed;
  public:
        void start (int s) {
                speed = s;
                cout << " Motor started at speed:
                        " << speed << endl;
        }
        void stop () {
                Sped () {
                cout << " motor stopped
                        endl;
        }
};
```

Exit Ticket: oop improves codereuability & modularity in robotics.

UNIT-4

Task A:

Q.1) What is latency in robotics?

Ans⇒ It is the delay between input (sensor data) & output (robot action).

Q.2) Why deterministic timing is imported?

Ans⇒ It ensures tasks execute at predictable intervals, which is critical for safe robot operation.

Q.3) What is a control loop?

Ans⇒ It continuously reads sensor data, processes it, and sends commands to activators.

Task B:

<Q>

Q.1) Is this system safe? Why or why not?

Ans⇒ No, the system is not safe because it misses deadlines, leading to unpredictable robot behaviour.

Q.2) What could be the reason for delay?

Ans⇒
- Heavy computations
- Hardware limitations
- Poor scheduling
- Blocking operations

Task C:

START
Loop forever:
read sensor data
process data & make decision
Send command to motor
Wait until next 50ms cycle.
END Loop.

Exit Ticket: Missing a control loop deadline can cause unsafe or incorrect robot action.